

Drugs for Disorders of the Respiratory System

By Banchayehu Firehun (B.Pharm, MSc.)

Asthma

- Asthma is a chronic inflammatory disease characterized by
 - Inflammation of the airways
 - Bronchial hyper-reactivity
 - Bronchospasm
- Major symptoms
 - Shortness of breath
 - Cough
 - Chest tightness
 - Wheezing
 - Rapid respiration

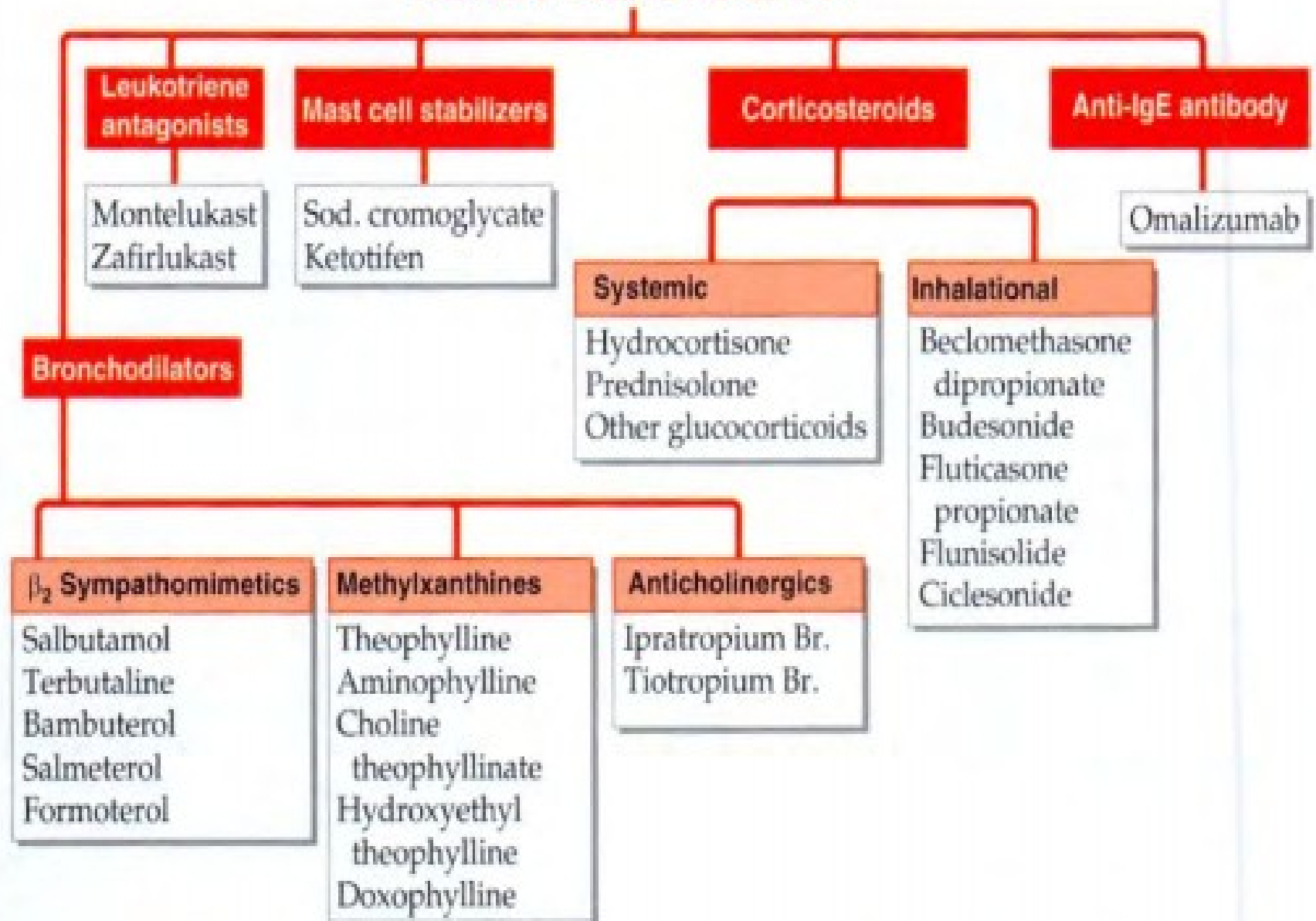
TABLE 76.5 ■ Classification of Asthma Severity and Recommended Step for Initial Treatment

		Persistent Asthma		
	Intermittent Asthma	Mild	Moderate	Severe
CURRENT IMPAIRMENT				
Asthma symptoms	≤2 days/week	>2 days/week, but not daily	Daily	Throughout the day
Nighttime awakenings	≤2 times/month	3–4 times/month	More than once a week, but not nightly	Often 7 times/week
SABA used to control symptoms (but not to prevent EIB)	≤2 days/week	>2 days/week, but not daily, and not more than once on any day	Daily	Several times a day
Effect on normal activity	None	Minor limitation	Some limitation	Severe limitation
Lung function tests	Normal FEV ₁ between exacerbations FEV ₁ > 80% of predicted FEV ₁ /FVC normal ^a	FEV ₁ > 80% of predicted FEV ₁ /FVC normal ^a	FEV ₁ >60% but <80% of predicted FEV ₁ /FVC reduced 5% ^a	FEV ₁ <60% of predicted FEV ₁ /FVC reduced >5% ^a
FUTURE RISK				
Exacerbations requiring oral glucocorticoids	0–1/yr	≥2/yr	≥2/yr	≥2/yr
Recommended step for initial treatment ^b	Step 1	Step 2	Step 3	Step 4 or 5

Treatment of asthma

- **Goals of therapy**
 - To decrease the intensity and frequency of asthma symptoms
 - To prevent future exacerbations
 - To minimize limitations in activity related to asthma symptoms
- **Drugs**
 - Anti-inflammatory drugs
 - Bronchodilators
 - Most of them are administered by inhalation

DRUGS FOR BRONCHIAL ASTHMA



Bronchodilators

- Provide symptomatic relief in patients with asthma and COPD
- **MoA:** cause bronchodilatation through $\beta 2$ -receptor stimulation
 - Increased cAMP formation in bronchial muscle cells
 - Relaxation
- Have no anti-inflammatory effects

β 2-agonists

- **Quick relief**
 - Short-acting β 2 agonists (SABAs)
 - Have a rapid onset of action (5 to 30 minutes)
 - Taken as PRN
 - Provide relief for 4 to 6 hours
 - Appropriate for patients with mild, intermittent asthma or exercise-induced bronchospasm
 - Not be used as monotherapy for patients with persistent asthma
 - Includes **albuterol (salbutamol), levalbuterol & terbutaline**

β2-agonists

- **Long-term control**
 - Long-acting β2 agonists (LABAs)
 - Providing bronchodilation for at least 12 hours
 - Taken as a fixed schedule
 - Should be used only in combination with an asthma controller medication, such as an inhaled corticosteroid (ICS).
 - Used as adjunctive therapy for attaining control in moderate to severe asthma
 - Include **Salmeterol** and **formoterol**
- Ultralongacting β2 agonists, such as **indacaterol**, **olodaterol**, **vilanterol**, and **bambuterol**.
 - Mostly for COPD

β 2-agonists

- Adverse effects:
 - Tachycardia/ angina
 - Hyperglycemia
 - Hypokalemia
 - Hypomagnesemia
 - These AEs are minimized with inhaled

Anti-cholinergic drugs

- Mostly used for COPD
- Ipratropium (short-acting, 4-6 hrs) and tiotropium (long acting, 24 hrs)
- Ipratropium is effective against allergen-induced asthma and EIB,
 - but is less effective than the beta2 agonists
- Tiotropium is approved for maintenance therapy of bronchospasm associated with COPD
- MoA: block muscarinic cholinergic receptors in the bronchi
 - Prevent bronchoconstriction and mucus secretion
- **Adverse effects:** xerostomia and bitter taste

Methylxanthines: Theophylline

- **MoA** of methylxanthines
 - Release of Ca^{2+} from sarcoplasmic reticulum, especially in skeletal and cardiac muscle
 - Inhibition of phosphodiesterase (PDE) which degrades cyclic nucleotides
 - concentration of cyclic nucleotides is increased
 - Bronchodilatation in bronchial muscle
 - Blockade of adenosine receptors

Pharmacokinetics

- Well absorbed orally
- Distributed in all tissues and 50% plasma protein bound
- Extensively metabolized in liver by demethylation and oxidation primarily by CYP1A2
- Only 10% is excreted unchanged in urine
- The $t_{1/2}$ in adults is 7- 12 hours
- Children eliminate it much faster ($t_{1/2}$ 3-5 hours) and elderly more slowly

Adverse effects

- Theophylline has a narrow margin of safety
- Headache, nervousness and nausea
- Cardiac arrhythmias and seizures
- Drug-drug interaction
 - Drugs enhance theophylline metabolism: phenytoin, rifampicin, phenobarbitone
 - Drugs inhibit theophylline metabolism: erythromycin, ciprofloxacin, cimetidine

Corticosteroids/Glucocorticoids

- Glucocorticoids are the drugs of choice for long-term control of airway inflammation
- Administered usually by **inhalation**, but may also be IV or oral
- Severe persistent asthma may require the addition of a short course of oral or intravenous corticosteroids
- Use in Asthma
 - prophylaxis of chronic asthma
 - persistent asthma

Glucocorticoids

- **MoA:** Inhibit the release of arachidonic acid through inhibition of phospholipase A2
 - thereby producing direct anti-inflammatory properties in the airways
 - **Decreasing** the inflammatory cascade (eosinophils, macrophages, and leukocytes)
 - Decreased edema of the airway mucosa
 - Decrease airway mucus production
 - may increase the number of bronchial beta2 receptors, as well as their responsiveness to beta2 agonists

Glucocorticoids

- Inhaled glucocorticoids (IGC): Beclomethasone, budesonide, fluticasone, mometasone, triamcinolone
- Oral CS: Methylprednisolone, prednisone, and prednisolone
- Severe exacerbation of asthma (status asthmaticus) may require intravenous **methylprednisolone** or oral **prednisone** to reduce airway inflammation.

Glucocorticoids

- Adverse effects
 - Oropharyngeal candidiasis
 - Dysphonia
 - Sore throat
 - Adrenal suppression (high dose, long term use)
 - Slow growth in children and adolescents
 - Bone loss (long term use)
 - Hyperglycemia and peptic ulcer disease (especially in oral)
 - ❖ Inhaled steroids are safe during pregnancy

Mast cell stabilizers

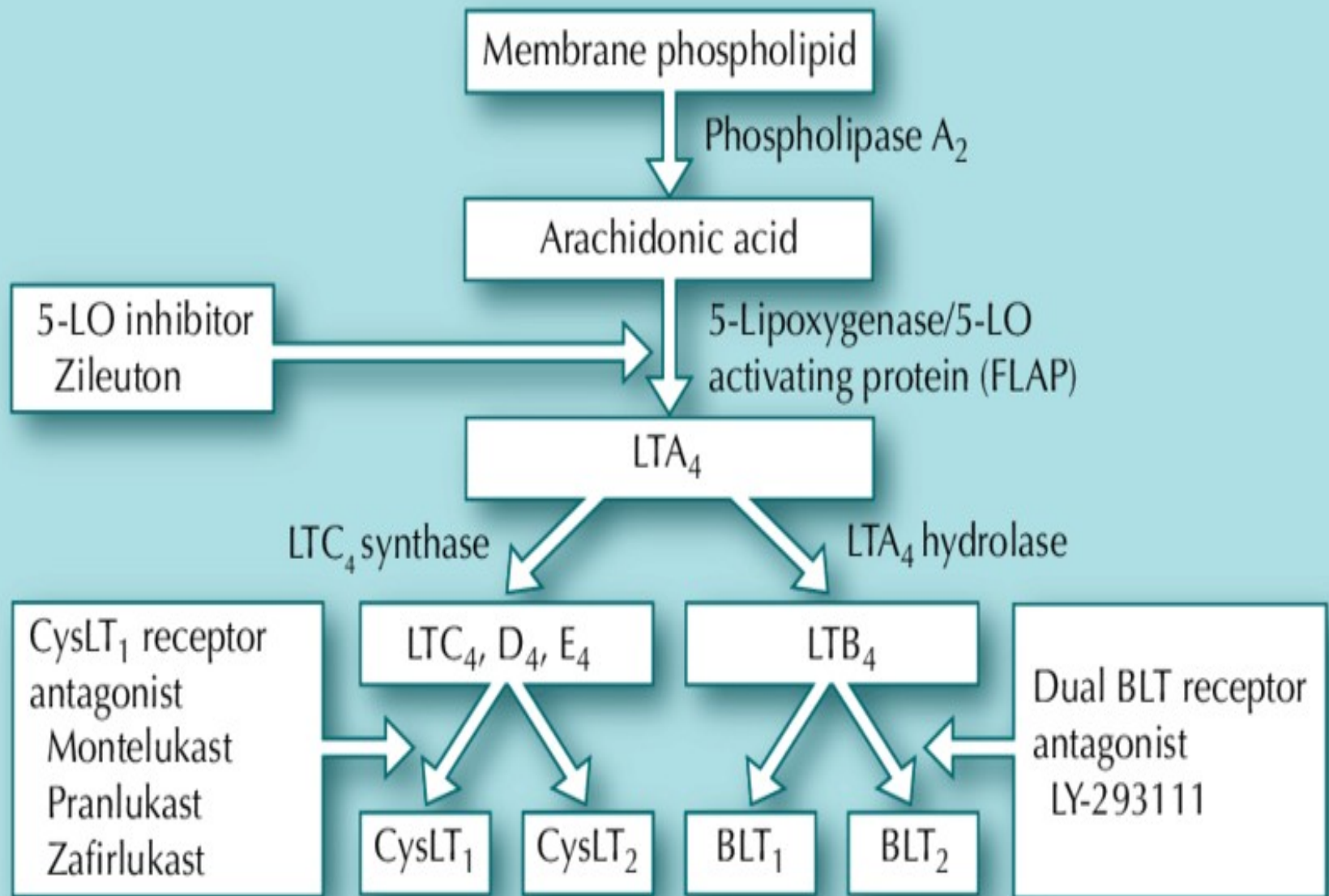
- **Sodium cromoglycate (Cromolyn sod.)**
 - Prophylactic anti-inflammatory agent
 - **Inhibits mast cell degranulation** and release of **histamine, LTs, ILs...**
 - Is **not** a bronchodilator and does not antagonize constrictor action of histamine, Ach, or LTs
 - Therapeutic uses
 - Alternative therapy
 - Exercise-induced bronchospasm
 - Allergic rhinitis

Mast cell stabilizers

- Pharmacokinetics
 - Not absorbed orally
 - Available as a nebulized solution
 - Has short duration of action
 - Rapidly excreted unchanged in urine and bile
- Adverse effects
 - Cough, throat irritation, and unpleasant taste

Leukotriene modifiers

- **Zileuton**
 - Inhibitor of **5-lipoxygenase (5-LOX)**
 - Preventing the formation of both LTB₄ and the cysteinyl leukotrienes
- **Zafirlukast and montelukast**
 - Selective antagonist of the cysteinyl leukotriene-1(**cysLT1**) receptor
 - Used for prophylactic therapy of mild-to-moderate asthma as alternatives to inhaled glucocorticoids.
 - In severe asthma, they have additive effect with inhaled steroids, may permit reduction in steroid dose and need for rescue **β2**-agonist inhalations.
 - Effective in **aspirin-induced** asthma and **exercise** induced asthma.



Leukotriene modifiers

- **Pharmacokinetics**
 - Orally active and highly protein bound.
 - Undergo extensive hepatic metabolism
 - Food reduces zafirlukast absorption
 - The plasma $t_{1/2}$ of montelukast is 3-6 hours, while that of zafirlukast is 8-12 hours, & zileuton is 2.5 hours.
 - **Zileuton** and its metabolites are excreted in urine
 - **Zafirlukast, montelukast**, and their metabolites undergo biliary excretion

Leukotriene modifiers

- **Adverse effects**
 - **Neuropsychiatric** effects (depression, anxiety, abnormal dreams, hallucinations, insomnia, and suicidal thinking)
 - **Zafirlukast**: headache and GI disturbance
 - Zileuton: hepatotoxic and inhibitor of CYP1A2
 - Zafirlukast is an inhibitor of CYP2C8, 2C9, and 3A4

Monoclonal antibodies

- Omalizumab
 - Selectively binds to human immunoglobulin E (IgE)
 - Decreased binding of IgE to its receptor on the surface of mast cells and basophils
 - No benefit has been noted in non-allergic asthma
 - Reserved for resistant asthma patients with positive skin tests or raised IgE levels
- Mepolizumab, benralizumab, and reslizumab
 - Are interleukin-5 (IL-5) antagonists

Monoclonal antibodies

- Indicated for the treatment of severe persistent asthma in patients who are poorly controlled with conventional therapy
- Their use is limited by the:
 - High cost
 - Route of administration (IV for reslizumab and SC for others)
 - Adverse effect profile
- Adverse effects:
 - Anaphylactic reaction (rare)
 - Arthralgias, fever, rash
 - Increased risk of infections

TABLE 76.11 ■ Stepwise Approach to Managing Asthma

	Long-Term Control Drugs (Taken Daily)		Quick-Relief Drugs (Taken PRN)
	Preferred	Alternative	
ADULTS AND CHILDREN AGE 12 AND OLDER			
Step 1	No daily medication needed		SABA
Step 2	Low-dose IGC	Cromolyn, LTRA, or theophylline	SABA
Step 3	Low-dose IGC + LABA OR Medium-dose IGC	Low-dose IGC + either LTRA, theophylline, or zileuton	SABA
Step 4	Medium-dose IGC + LABA	Medium-dose IGC + either LTRA, theophylline, or zileuton	SABA
Step 5	High-dose IGC + LABA		SABA
Step 6	High-dose IGC + LABA + oral glucocorticoid		SABA
CHILDREN AGE 5–11 YEARS			
Step 1	No daily medication needed		SABA
Step 2	Low-dose IGC	Cromolyn, LTRA, nedocromil, or theophylline	SABA
Step 3	Low-dose IGC + either LABA, LTRA, or theophylline OR Medium-dose IGC		SABA
Step 4	Medium-dose IGC + LABA	Medium-dose IGC + either LTRA or theophylline	SABA
Step 5	High-dose IGC + LABA	High-dose IGC + either LTRA or theophylline	SABA
Step 6	High-dose IGC + LABA + oral glucocorticoid	High-dose IGC + either LTRA or theophylline + oral glucocorticoid	SABA
CHILDREN AGE 0–4 YEARS			
Step 1	No daily medication needed		SABA
Step 2	Low-dose IGC	Cromolyn or montelukast	SABA
Step 3	Medium-dose IGC		SABA
Step 4	Medium-dose IGC + either LABA or montelukast		SABA
Step 5	High-dose IGC + either LABA or montelukast		SABA
Step 6	High-dose IGC + either LABA or montelukast (Low-dose oral glucocorticoids, if needed.)		SABA

IGC, Inhaled glucocorticoid; LABA, long-acting beta₂ agonist; LTRA, leukotriene receptor antagonist; SABA, short-acting beta₂ agonist.

Pharmacotherapy of cough and nasal decongestant

- Cough is a protective reflex, its purpose being expulsion of respiratory secretions and foreign particles from air passages.
- It may be productive or non-productive
- Before treating cough, identification of its cause is important.

Pharmacotherapy of cough

- **Expectorants**
 - Increase bronchial secretion or reduce its viscosity, facilitating its removal by coughing.
 - Example: **Guaifenesin**
- **Mucolytics**
 - Reacts directly with mucus to make it less viscous
 - Eg. hypertonic saline and acetylcysteine
- **Antitussives**
 - Suppress cough centrally and peripherally
 - Should be used **only** for dry **non-productive** cough
 - Classified as opioid and non-opioid antitussives

Opioid antitussives

- **Codeine**
 - **Decreases** the sensitivity of cough center in the CNS to peripheral stimuli and decrease the frequency and intensity of cough
 - Its therapeutic effects occur at doses lower than those required for analgesia.
 - At higher doses respiratory depression and drowsiness can occur, especially in **children**
 - Adverse effects: constipation, dysphoria, fatigue, & **addiction**.

Opioid antitussives

- **Pholcodine**
 - It has practically no analgesic or addicting property
 - Similar in efficacy as antitussive to codeine
 - Longer acting-acts for 12 hours

Nonopioid antitussives

- **Dextromethorphan**
 - A synthetic central N-methyl D-aspartate (NMDA) receptor antagonist
 - Does not depress mucociliary function of the airway mucosa and is practically devoid of constipating action.
 - **Adverse effects:** dizziness, nausea, drowsiness; at high doses hallucinations and ataxia may occur.

Adjuvant antitussives

- **Antihistamines**
 - Added to antitussive/expectorant formulations
 - First-generation antihistamines are commonly used
 - Chlorpheniramine
 - Diphenhydramine
 - Promethazine
- **Bronchodilators:** Salbutamol and Terbutaline

Drugs used to treat allergic rhinitis

- Allergic rhinitis is an inflammatory disorder that affects the upper airway.
- Major symptoms are sneezing, rhinorrhea (runny nose), pruritus (itching), and nasal congestion
 - caused by dilation and increased permeability of nasal blood vessels.
- Allergic rhinitis has two major forms: seasonal and perennial
- Treatment includes:
 - Antihistamines
 - Corticosteroids
 - α -Adrenergic agonists

Antihistamines

- H1 receptor antagonists
 - The second-generation antihistamines (**fexofenadine**, **loratadine**, desloratadine, cetirizine) are mostly used.
 - Topical intranasal antihistamines include olopatadine and azelastine.
 - Intranasal antihistamines provide increased delivery of the drug with fewer adverse effects.

Corticosteroids

- Intranasal corticosteroids, such as beclomethasone, budesonide, fluticasone, ciclesonide, mometasone, and triamcinolone.
- Onset of action that ranges from 3 to 36 hours.
- For patients with chronic rhinitis, improvement may not be seen until 1 to 2 weeks after starting therapy.
- Adverse effects: nasal irritation, nose bleed, sore throat, and candidiasis (rarely).

α -Adrenergic agonists

- Nasal decongestants
- Activate α_1 -adrenoceptor in the arterioles of nasal mucosa
 - Vasoconstriction, which leads to decreased edema and reduce airway resistance.
- Includes: phenylephrine, pseudoephedrine and oxymetazoline
- Intranasal formulations should be used for no longer than 3 days due to the risk of **rebound** nasal congestion
- Oral formulations results in a longer duration of action but also increased systemic effects, such as increased blood pressure and heart rate, excitation, & **abuse (pseudoephedrine)**

Thank you!